Weekly Test

SECONDARY COURSE - MATHS 211

Roll No. | | | | | | | | | | | | |

Code No.WT1/X/211/26
Set A

Day and Date of Examination ______________________

Signature of Invigilat1ors 1. ______________________

2. ______________________

General Instructions :

1. Candidate must write his/her Roll Number on the first page of the Question Paper.
2. Please check the Question Paper to verify that the total pages and the total number of questions
contained in the Question Paper are the same as those printed on the top of the first page. Also check to see that the questions are in sequential order.
3. For the objective-type of questions, you have to choose any one of the four alternatives given in the question i.e. (A), (B), (C) or (D) and indicate your correct answer in the Answer-Book given to you.
4. All the questions including objective-type questions are to be answered within the allotted time and no separate time limit is fixed for answering objective-type questions.
5. Making any identification mark in the Answer-Book or writing Roll Number anywhere other than the specified places will lead to disqualification of the candidate.
6. In case of any doubt or confusion in the question paper, the English Version will prevail.
7. Write your Question Paper Code No. 70/OS/2, Set A on the Answer-Book.
8. (a) The Question Paper is in English/Hindi medium only. However, if you wish, you can answer in any one of the languages listed below:
English, Hindi, Urdu, Punjabi, Bengali, Tamil, Malayalam, Kannada, Telugu, Marathi, Odia, Gujarati, Konkani, Manipuri, Assamese, Nepali, Kashmiri, Sanskrit and Sindhi.
You are required to indicate the language you have chosen to answer in the box provided in the Answer-Book.

(b) If you choose to write the answer in the language other than Hindi and English, the responsibility for any errors/mistakes in understanding the questions will be yours only.

General Instruction :

1. Answers of all questions are to be given in the Answer-Book given to you.
2. 15 minutes time has been allotted to read this Question Paper. The question paper will
be distributed at 10.30 a.m. From 10.30 a.m. to 10.45 a.m., the students will read the question paper only and will not write any answer on the Answer-Book during this period.

Note :

(i) This question paper consists of 25 questions in all.
(ii) All questions are compulsory.
(iii) Marks are given against each question.
(iv) Section – A consists of:
(a) Q. No. 1 to 10 – Multiple Choice type Questions (MCQs) carrying 1 mark each. Select and write the most appropriate option out of the four options given in each of these questions.
(b) Q. No. 11 to 15 – Objective type questions. Q. No. 11 to 15 carry 2 marks each (with 2 sub-parts of 1 mark each) and Q. No. 16 carries 5 marks (with 5 sub-parts of 1 mark each). Attempt these questions as per the instructions given for each of the questions 11 to 15.
(v) Section – B consists of:
(a) Q. No. 17 to 21 – Very Short Answer type questions carrying 2 marks each.
(b) Q. No. 22 to 23 – Short Answer type questions carrying 3 marks each.
(c) Q. No. 24 – Long Answer type questions carrying 5 marks each.

Time : 2 Hours **Maximum Marks:** 50

SECTION A

1. If $120^2-20^2=25p$, then p is equal to: 1

(A) 16

(B) 140

(C) 560

(D) 14000

2. The HCF of $36a^5b^2$ and $90a^3b^4$ is: 1

(A) $36a^3b^2$

(B) $18a^3b^2$

(C) $90a^3b^4$

(D) $180a^5b^4$

3. Which of the below options is correct 1

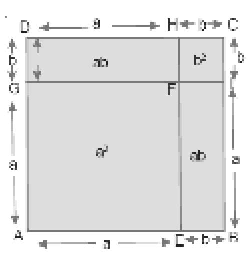

Fig A

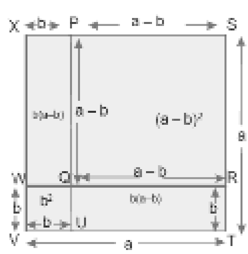

Fig B

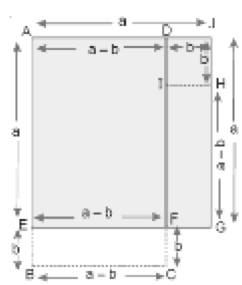

Fig C

(A) Fig A = $(a + b)^2 = a^2 + 2ab + b^2$, Fig B = $(a - b)^2 = a^2 - 2ab + b^2$, Fig C = $(a + b)(a - b) = a^2 - b^2$

(B) Fig A = $(a - b)^2 = a^2 - 2ab + b^2$, Fig B = $(a + b)^2 = a^2 + 2ab + b^2$, Fig C = $(a + b)(a - b) = a^2 - b^2$

(C) Fig A = $(a - b)^2 = a^2 - 2ab + b^2$, Fig B = $(a + b)(a - b) = a^2 - b^2$, Fig C = $(a + b)^2 = a^2 + 2ab + b^2$

(D) Fig A = $(a + b)^2 = a^2 + 2ab + b^2$, Fig B = $(a + b)(a - b) = a^2 - b^2$, Fig C = $(a - b)^2 = a^2 - 2ab + b^2$

4. Which of the following are linear equations in one variable? 1

(i) $2x + 5 = 8$
(ii) $3y - z = y + 5$
(iii) $x^2 - 2x = x + 3$
(iv) $3x - 7 = 2x + 3$
(v) $2 + 4 = 5 + 1$

(A) (ii)&(iii)

(B) (i)&(iv)

(C) (iv)&(v)

(D) (ii)&(v)

5. Which of the following quadratic equations are in standard form? 1

(i) $3y^2 - 2 = y + 1$
(ii) $5 - 3x - 2x^2 = 0$
(iii) $(3t - 1)(3t + 1) = 0$
(iv) $5 - x = 3x^2$

(A) (ii)&(iii)

(B) (i)&(iv)

(C) (iv)&(iii)

(D) All are in non-standard form

6. We have $ax^2 + bx + c = 0$ where a, b and c are constants and x is a variable then x the solutions or roots of this quadratic equation is 1

(A) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{4ac}$
(B) $x = \frac{-b \pm \sqrt{b^2 - 2ac}}{4a}$
(C) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
(D) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{4a}$

7. In an A.P., the sum of three consecutive numbers is 18 and their product is 120, then the three numbers are: 1

(A) 3, 4, 11
(B) 2, 6, 10

(C) 1, 7, 10

(D) 3, 6, 9

8. The sum of an AP whose first term is a and last term is l and number of terms is n is given by 1

(A) $S_n = \frac{n}{2}[2a + (n-1)d]$
(B) $S_n = a + (n+1)d$
(C) $S_n = \frac{n}{2}(a+l)$
(D) $S_n = \frac{n}{2}(an-2l)$

9. The product of $(2a+3b)^2$ is: 1

(A) $4a^2 + 12ab + 9b^2$
(B) $4a^2 - 9b^2$
(C) $4a^2 - 12ab + 9b^2$
(D) $9a^2 + 12ab + 4b^2$

10. If $5 + x = 8$ so, x is: 1

(A) 1
(B) 13
(C) 3
(D) -3

11. Write TRUE for correct statement and FALSE for incorrect statement

(i) Every quadratic equation has exactly one root. 1

(ii) If the quadratic equation ax2 + bx + c = 0 has equal roots, then b2 – 4ac = 0 1

(iii) For the A. P. 10, 5, 0, – 5, ..., the common difference is equal to 5. 1

(iv) The sequence 2, 2, 2, 2, 2, 2, ..., is an A. P. 1

12. Find the products of

(i) $\left(\frac{x}{3}+1\right)^2$ 1

(ii) $\left(x+\frac{4}{3}\right)\left(x+\frac{3}{4}\right)$ 1

13. Solve

(i) $7x+2=8$ 1

(ii) $\frac{3y}{2}-3=9$ 1

14. Without determining the roots, comment on nature of roots of following equations:

(i) $3x^2+7x+2=0$ 1

(ii) $25x^2+20x+4=0$ 1

15. Fill in the blanks

(i) Sum of 2 + 5 + 8 + 11 + ... + 59 is __________ 1

(ii) Sum of all natural numbers between 1 and 1000 which are divisible by 7 is ___________ 1

(iii) nth term of 5, 9, 13, 17, ... is ________ 1

(iv) For a quadratic equation $ax^2+\mathrm{b}x+\mathrm{c}=0, \mathrm{a}\neq 0$ will not have any real roots when the Determinant $D=b^2+4\mathrm{ac}$ is _________ 1

16. Read the table carefully and answer the questions that follow

A teacher writes the following quadratic equations on the board and asks the students to identify the nature of their roots using the discriminant $D=b^2-4\mathrm{ac}$

Equation	Value of Discriminant (D)
P : $x^2-6x+9=0$	0
Q : $x^2+5x+6=0$	1
R : $x^2+4x+5=0$	−4
S : $2x^2+7x+3=0$	25

Based on the above table, answer the following questions.

(i) Which equation has equal and real roots? 1

(A) P

(B) Q

(C) R

(D) S

(ii) Which equation has no real roots? 1

(A) P

(B) Q

(C) R

(D) S

(iii) Which equation has two distinct real roots? 1

(A) P only

(B) Q and S

(C) R only

(D) P and R

(iv) The discriminant of equation S is: 1

(A) 0

(B) –25

(C) 25

(D) 49

(v) If the discriminant of a quadratic equation is negative, then its roots are: 1

(A) Equal and real

(B) Distinct and real

(C) Not real

(D) One real and one imaginary

17. Find the Cube of $(a+b)$ 2

18. The sum of three consecutive even integers is 36. Find the integers. 2

19. 2

The digit at ten's place of a two-digit number is 2 more than twice the digit at unit's place. If product of digits is 24, find the two digit number.

Or

The hypotenuse of a right-angled triangle is 13 cm. If the difference of remaining two sides is 7 cm, find the remaining two sides.

20. 2

Each term of an AP whose first term is a and common difference is d, is doubled. Is the resulting pattern an AP? If so, find its first term and common difference.

Or

The angles of a triangle are in AP. If the smallest angle is one-third the largest angle, find the angles of the triangle.

21. 2

Asha is five years older than Robert. Five years ago, Asha was twice as old as Robert was then. Find their present ages.

Or

Find three irrational numbers between 3 and 8.

22. The length of a rectangular garden is 7 m more than its breadth. If area of the garden is 144 m2, find the length and breadth of the garden. 3

23. Find the sum of all 3-digit numbers which leave the remainder 1, when divided by 4. 3

Or

If $m=\frac{x+1}{x-1}$ and $n=\frac{x-1}{x+1}$, find m^2+n^2-mn

24. 5

Had Ajay scored 10 more marks in his test out of 30 marks, 9 times these marks would have been the square of his actual marks. How many marks did he get in the test?

Or

Show that the sum of an AP whose first term is a, the second term is b and the last term is c, is equal to $\frac{(a+c)(b+c-2a)}{2(b-a)}$